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Leveraging Propargyl Sulfides as Radical Trappers for the Synthesis of Sulfonyl Allenes

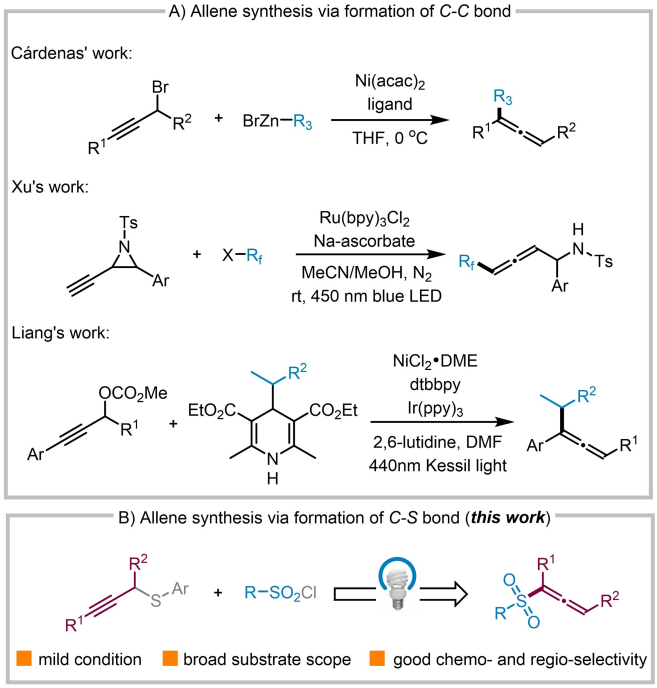
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This work showcases a novel visible-light mediated allenylation protocol employing propargyl sulfides as a pioneering radical trapping methodology. A diverse array of substituted sulfonyl allenes were successfully synthesized under mild conditions with commendable levels of chemo- and regio-selectivity. Mechanistically, this process entails the formation of sulfonyl radicals, their subsequent addition to alkynes, the subsequent elimination of S-centered radicals, and Cl-atom abstraction, thereby facilitating the propagation of this radical chain reaction.

Introduction

Allenes constitute a significant category of organic compounds characterized by distinctive twisted orthogonal π-systems.[1] They serve as pivotal structural units in numerous natural products and pharmaceuticals, showcasing their prominence in both medicinal chemistry and natural product synthesis. Additionally, they emerge as highly versatile synthons in contemporary organic synthesis methodologies, facilitating the construction of diverse molecular architectures and enabling efficient design strategies for bioactive molecules. A plethora of methodologies for the synthesis of allenes have been extensively documented in the literature. In pursuit of advancing increasingly reliable methodologies, there has been a rapid surge in the utilization of radical species for the synthesis of allenes. This radical approach has recently received significant attention and its potential has been convincingly demonstrated.[2] These include radical 1,4-addition to 1,3-enynes,[3] radical synthesis of allenes from propargylic compounds,[4] radical allene C(sp^2)-H functionalization,[5] and various other efficient synthetic routes.[6]

As an important complement to previous studies, the radical-mediated conversion of propargylic compounds into allenes has emerged as a versatile synthetic route in recent years (Scheme 1A). For instance, Cárdenas and coworkers observed that rather than yielding propargylic compounds, nickel-catalyzed the cross-coupling reaction between alkylzinc halides and propargyl bromides, provides a pathway to access allenes (Scheme 1A, top).[4a]



Scheme 1. Radical synthesis of allenes from propargylic compounds.

In 2020, Xu and Lan et al. reported a photocatalytic radical synthesis of functionalized allenes under visible-light irradiation (Scheme 1A, middle).[4b] This method utilizes readily accessible terminal alkynyl aziridines. In 2021, the Liang group disclosed a photoredox/nickel dual-catalyzed alkylation approach employing propargylic carbonates and alkyl 1,4-dihydropyridine derivatives (1,4-DHPs) serving as alkyl radical precursors (Scheme 1A, bottom).[4c] Liu et al. pioneered a novel approach for the direct synthesis of chiral allenes through enantioselective copper-catalyzed cyanation of propargylic C–H bonds.[4d] Studer and colleagues reported a example of radical allylation employing a homoallylic alcohol, albeit yielding modestly.[7] In 2021, Zhu and coworkers disclosed the radical chain isomerization of *N*-

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sulfonylynamides to ketenimines via the addition of sulfonyl radicals to ynamides.[8] As a result, radical synthesis of allenes from propargylic compounds via C–C bond formation has been extensively explored. However, to the best of our knowledge, radical synthesis of allenes from propargylic compounds via C-heteroatom bond formation has been scarcely documented.

Motivated by our interest in radical-mediated allene synthesis,[9] we postulated that the addition of a heteroatom-centered radical to propargylic compounds could lead to the formation of corresponding allenes via C-heteroatom bond formation under mild conditions. To achieve this transformation, several challenging issues should be addressed: (1) Considering the high reactivity of the allene products, strategies need to be devised to prevent further transformations of the products.[10] (2) When employing internal alkynes, methods for controlling site-selectivity during the radical addition process are crucial. (3) Mechanistically, the generation of thiyl radicals would be generated. Strategies should be developed to prevent their addition to the alkyne substrate or the allene product.

Herein, we present experimental evidence supporting this hypothesis (Scheme 1B). Employing propargylic sulfides as substrates, the reaction proceeds through the addition of a sulfonyl radical to the alkyne followed by the elimination of an S-centered radical. Notably, this transformation operates under mild visible-light-mediated photocatalytic conditions and affords valuable sulfonyl allenes.

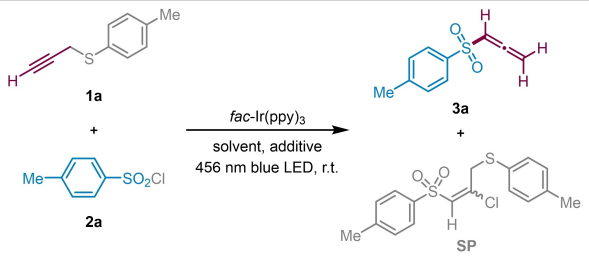
Results and Discussion

At the outset, propargylic sulfide (**1a**) was selected as the model substrate to undergo reaction with tosyl chloride (**2a**) as the precursor of tosyl radical under photocatalytic conditions. The reaction furnished the desired allene product (**3a**) utilizing *fac*-Ir(ppy)$_3$ as the photocatalyst. Several parameters, such as solvent, and additive, were systematically examined to optimize the reaction conditions (Table 1). Taking into account that H_2O may aid in the dissolution of the inorganic base, we firstly conducted optimization of mixed solvents (entries 1–6). It was observed that the utilization of $CHCl_3/H_2O$ resulted in a higher yield (39%, entry 2). A range of bases, such as $KHCO_3$, 4-Dimethylaminopyridine (DMAP), and Et_3N, were subsequently evaluated (entries 7–10), revealing that exclusion of the base led to an improved yield (entry 10). However, the reaction performed in the absence of H_2O resulted in a decreased yield (entry 11). Intriguingly, reducing the reaction concentration markedly enhanced the yield (entry 12). The conversion of substrate (**2a**) reached approximately 95%. A side product, alkene (**SP**), was formed with a yield of 30% via the atom transfer radical addition (ATRA) process. Furthermore, control experiments demonstrated the essential roles of the photocatalyst and blue light irradiation in the reaction (entries 13–14).

With the optimized conditions established for the synthesis of sulfonyl allenes, the compatibility of various propargylic sulfides and sulfonyl chlorides was investigated (Scheme 2). Common functional groups on the phenyl ring, such as ether (**3b**), benzene (**3f**), nitrile (**3g**), ketone (**3h**), halides (**3i**--**3k**), nitro group (**3o**), and trifluoromethyl group (**3p**), exhibited compatibility under the reaction conditions. Regardless of the presence of electron-withdrawing or -donating substituents at the *para*-position of the phenyl ring, the reactions proceeded smoothly to afford synthetically valuable yields (**3a**–**3k**). Phenylsulfonyl chlorides bearing *ortho*-, *meta*- or di-substituents underwent facile allenylation (**3l**--**3r**), albeit the *ortho*-methyl substituent resulted in a relatively lower yield (**3m**). Substituting the phenyl group with a naphthyl group (**3s**), thienyl group (**3u**), or a chromone moiety (**3t**) yielded the corresponding products with slightly reduced yields. It is noteworthy that reactions with alkyl sulfonyl chlorides efficiently furnished the desired allenes (**3v**--**3x**). Internal alkynes bearing with sulfide could also effectively trap the sulfonyl radical, affording 1,1-disubstituted allenes (**3y**) with excellent chemo- and regioselectivity. Additionally, employing secondary aryl sulfides bearing various alkyl moieties led to the formation of 1,3-disubstituted allenes (**3z**--**3ac**) in synthetically useful yields.

The scaled-up synthesis of **3a** utilizing flow chemistry techniques proceeded smoothly,[11] yielding a comparable result, thus underscoring the practicality of the protocol (Scheme 3A). To investigate the mechanistic pathway, we employed high-resolution mass spectrometry (HRMS) for reac-

Table 1. Reaction parameters survey.[a]



Entry	Solvent	Additive	Yield (%)[b]
1	$MeCN/H_2O = 10/1$	Cs_2CO_3	30
2	$CHCl_3/H_2O = 10/1$	Cs_2CO_3	39
3	$CH_2Cl_2/H_2O = 10/1$	Cs_2CO_3	36
4	$EtOAc/H_2O = 10/1$	Cs_2CO_3	27
5	$THF/H_2O = 10/1$	Cs_2CO_3	36
6	$DMSO/H_2O = 10/1$	Cs_2CO_3	n.r.
7	$CHCl_3/H_2O = 10/1$	$KHCO_3$	40
8	$CHCl_3/H_2O = 10/1$	DMAP	20
9	$CHCl_3/H_2O = 10/1$	Et_3N	34
10	$CHCl_3/H_2O = 10/1$	–	45
11	$CHCl_3$	–	24
12[c]	$CHCl_3/H_2O = 20/1$	–	66 (65)[d]
13[c,e]	$CHCl_3/H_2O = 20/1$	–	0
14[c,e]	$CHCl_3/H_2O = 20/1$	–	0

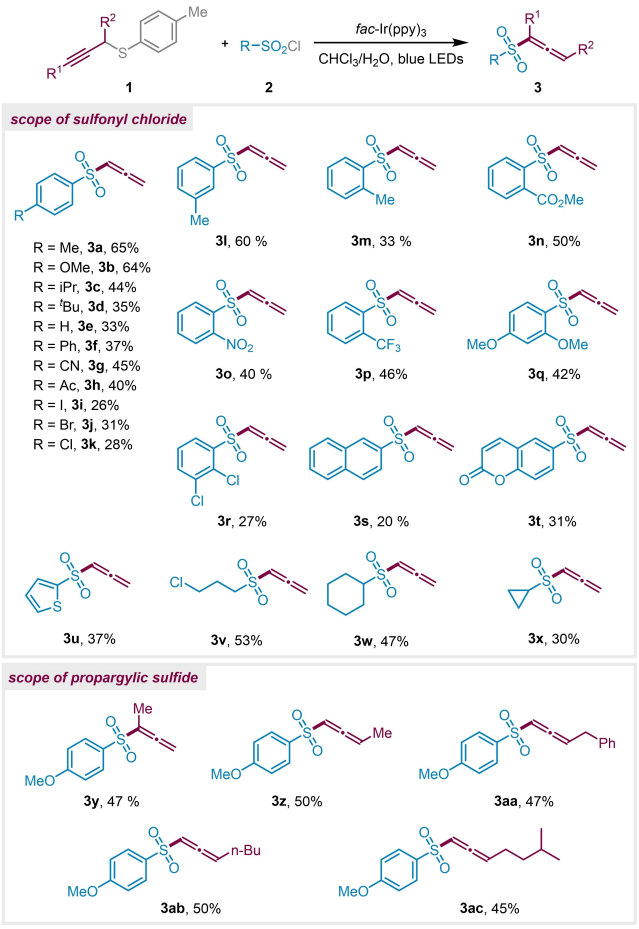
[a] Reaction conditions: **1a** (0.15 mmol, 1.5 equiv.), **2a** (0.1 mmol, 1.0 equiv.), Additive (0.1 mmol, 1 equiv.), *fac*-Ir(ppy)$_3$ (0.003 mmol, 0.03 equiv.), solvent (1.1 mL) was irradiated with 18 W 456 nm blue LEDs at r.t. for 4 h under N_2. [b] Yields of NMR products. [c] solvent (2.1 mL). [d] Isolated yields. [e] in the absence of photocatalyst. [f] in dark.

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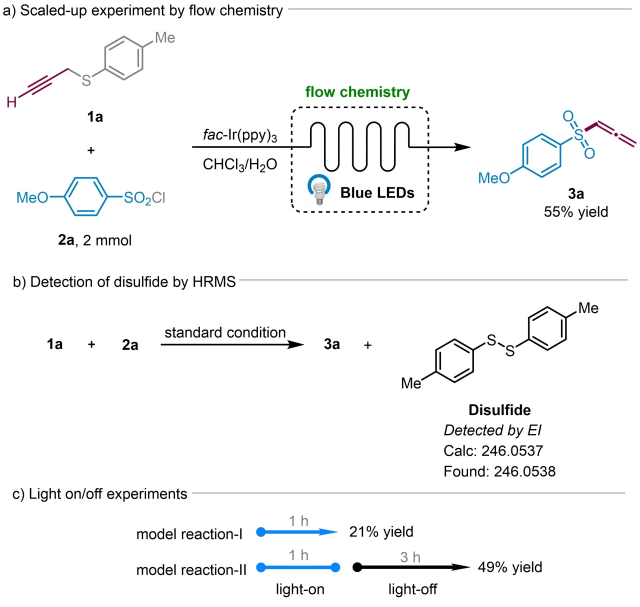
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Scheme 2. Substrate scope. Radical synthesis of allenes from propargylic compounds. Reaction conditions: **1** (0.15 mmol, 1.5 equiv.), **2** (0.1 mmol, 1.0 equiv.), *fac*-Ir(ppy)$_3$ (0.003 mmol, 0.03 equiv.), $CHCl_3/H_2O = 20/1$ (2.1 mL) were irradiated with 18 W 456 nm blue LEDs at r.t. for 4 h under N_2. Yield of isolated products was given.



Scheme 3. Mechanistic studies.

tion detection, revealing the presence of disulfide (Scheme 3B). Subsequently, light on/off experiments were performed on the model reaction, demonstrating that the reaction yield increased even when light was turned off (Scheme 3C). This observation suggests that the reaction operates via a radical chain process.

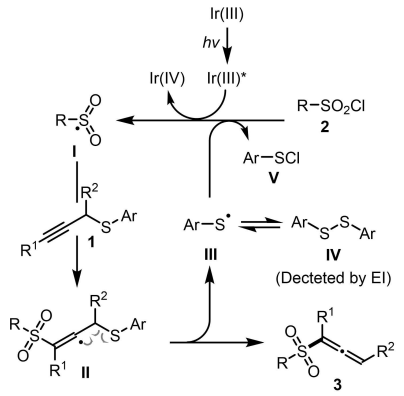
Based on these findings and previous literatures, a plausible mechanism for this radical-chain process is proposed in Scheme 4. Initially, the reaction is initiated by single-electron transfer (SET) between the excited Ir(III)* species and sulfonyl chloride (**2**). This results in the formation of the sulfonyl radical (**I**), which subsequently adds to alkyne (**1**) to generate the vinyl radical (**II**).[12] The subsequent elimination of the thiyl radical (**III**) occurs, leading to the formation of product (**3**). Dimerization of the thiyl radical (**III**) yields disulfide (**IV**). Concurrently, the generation of the sulfonyl radical is facilitated via Cl-atom transfer mediated by the thiyl radical (**III**),[13] thus perpetuating the radical chain process.

Conclusions

We present a novel photocatalytic allenylation protocol utilizing propargyl sulfides as a new radical trapping agent. The radical-chain mechanism involves sequential steps including addition of a sulfonyl radical to the alkyne, elimination of a thiyl radical, and chloride atom transfer. A diverse array of sulfonyl radicals readily undergo radical trapping by a range of propargyl sulfides. This methodology offers an efficient and gentle means for the synthesis of valuable sulfonyl allenes.

Experimental Section

To a reaction flask was added **1** (2.0 equiv.), sulfonyl chloride (0.1 mmol) and *fac*-Ir(ppy)$_3$ (3 mol%) under N_2, then $CHCl_3$ (2.0 mL) and H_2O (0.1 mL) was added to the mixture via syringe and the mixture was stirred at room temperature and under 18 W blue LEDs irradiation for 4 h. The reaction was extracted with ethyl acetate, and the combined organic layer was washed with brine, dried over $MgSO_4$, filtered, concentrated in vacuum, and purified by flash column chromatography on silica gel to give product **3**.



Scheme 4. Proposed Mechanism.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Radical reactions · Photocatalysis · Allene synthesis · Sulfides · Sulfonyl chlorides

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